HuggingFaceFW

FinePhrase

The Synthetic Data Playbook

Generating Trillions of the Finest Tokens with DataTrove

1.35B Samples

486B Tokens

4 Prompt Families

Contents

01

Project Overview

What is FinePhrase?

Massive Synthetic Dataset

FinePhrase transforms educational web content from FineWeb-Edu into structured formats using AI. It represents one of the largest synthetic data generation projects for LLM training.

Four Transformation Types

Content is transformed into FAQ, Math problems, Structured Tables, and Tutorials—each serving different training objectives for language models.

Powered by DataTrove

Built on Hugging Face's DataTrove pipeline library, enabling scalable, platform-agnostic data processing from loc al machines to SLURM clusters.

Generation Scale

Source Documents
339.3M

Output Samples
1.35B

Completion Tokens
486.4B

4x Expansion Rate

02

The FineData Ecosystem

FinePhrase within the HuggingFaceFW Family

🍷 FineData Team
Hugging Face Science Team

FineWeb
15T tokens English dataset for LLM pre-training
🍷 15 Trillion

FineWeb-Edu
Filtered educational content from FineWeb
📚 3.5 Billion

FineWeb2
Extension to 1000+ languages
🥂 1000+ Langs

FinePDFs
3T tokens from PDF documents
📄 3 Trillion

FineWiki
Updated Wikipedia in 300+ languages
🌐 300+ Langs

FinePhrase
Synthetic transformation of educational content
💬 486B Tokens

FineWeb-Edu → FinePhrase → Synthetic Augmentation

03

DataTrove: The Processing Engine

Platform-Agnostic Pipeline Processing Library

What is DataTrove?

DataTrove is a Python library for **processing, filtering, and deduplicating** text data at very large scale. It provide s prebuilt processing blocks with a framework for custom functionality.

Low memory usage & multi-step design

Core Capabilities

Readers
Multiple formats

Filters
SOTA quality

Deduplication
MinHash, Exact Substr

Writers
Flexible output

Tokenization
Multiple tokenizers

Inference
LLM generation

Execution Environments

LocalPipelineExecutor
Multi-CPU local execution

SlurmPipelineExecutor
SLURM cluster deployment

RayPipelineExecutor
Ray distributed compute

latform-agnostic: Same pipeline runs anyw here

```
pip install datatrove[inference]
```

04

Generation Model

SmolLM2-1.7B-Instruct

Model Selection

HuggingFaceTB/SmolLM2-1.7B-Instruct

SmolLM2-1.7B-Instruct was chosen for its optimal balance of efficiency and capability. Despite its compact size, it delivers high-quality synthetic content across diverse transformation tasks.

Parameters	Architecture
1.7B	Transformer

Why This Model?

Efficient Inference

Smaller model enables faster generation at scale, reducing computational costs for 486B tokens.

Instruction-Tuned

Fine-tuned for following complex prompts—critical for FAQ, Math, Table, and Tutorial generation.

Quality Output

Produces coherent, structured content matching the educational quality of FineWeb-Edu source.

Model Capabilities

FAQ Generation

Extracts key questions and provides clear answer s

Math Problems

Creates word problems with step-by-step solutio ns

Structured Tables

Organizes data into markdown tables

05

Generation Configuration

Parameters & Decoding Strategy

Core Parameters

Temperature
1.0
Creativity control

Top-p (Nucleus)
1.0
Full nucleus sampling

Top-k
50
Top-k sampling limit

Max Tokens
2048
Output length limit

Speculative Decoding

Method
Suffix

Speculative Tokens
32

Context & System Settings

Setting	Description	Value
Model Max Context	Total context window size	8192
System Prompt	Global instruction	None
Input Column	Source text field	text

Why These Settings?

1. **Temp=1.0:** Balanced creativity—neither too r andom nor too deterministic
2. **Top-p=1.0:** Full nucleus sampling for diverse outputs
3. **Top-k=50:** Limits vocabulary to top 50 toke ns per step
4. **Speculative:** Speeds up generation by 2-3x

06

Source Dataset

FineWeb-Edu Sample-350BT

FineWeb-Edu

Educational Content Filter

FineWeb-Edu is a filtered subset of FineWeb containing the most educational content. It uses a classifier to identify high-quality educational documents from web data.

Full Dataset	Sample Used
3.5B documents	350BT config

Input Statistics

Dataset Name
HuggingFaceFW/fineweb-edu

Configuration
sample-350BT

Split
train

Why FineWeb-Edu?

High Quality
Classifier-filtered for educational value

Pre-cleaned
Already deduplicated and filtered

Diverse
Broad educational topics covered

Safe
PII and toxic content filtered

07

Prompt Family 1

FAQ Generation

FAQ Prompt Template

```
Rewrite the document as a comprehensive FAQ (Frequently Asked Questions). Extract or infer the key questions a reader would have about thi
s topic, then provide clear, direct answers. Order questions logically, from foundational to advanced, or by topic area. Each answer should be sel
f-contained and understandable without reference to other answers. Ensure the FAQ works as a standalone document. Output only the FAQ, no
thing else.
Document: [[DOCUMENT]]
```

Training Objective

Question Answering
Teaches models to answer questions accurately

Information Extraction
Identifies key points in documents

Structured Output
Generates organized Q&A format

Key Design Elements

1. **Extract & Infer** — Both explicit and implicit questions
2. **Logical Ordering** — Foundational to advanced
3. **Self-Contained** — No cross-references needed
4. **Standalone** — Works independently

Generation Stats

Documents	338.97M
Tokens	148.1B
Mean Tokens/Sample	436.96

08

Prompt Family 2

Mathematical Word Problems

Math Prompt Template

Rewrite the document to create a mathematical word problem based on the numerical data or relationships in the text. Provide a step-by-step s olution that shows the calculation process clearly. Create a problem that requires multi-step reasoning and basic arithmetic operations. It shoul d include the question followed by a detailed solution showing each calculation step. Output only the problem and solution, nothing else.
Document: [[DOCUMENT]]

Key Design Elements

1. **Numerical Focus** — Extracts quantitative data
2. **Step-by-Step** — Detailed solution process
3. **Multi-Step** — Complex reasoning required
4. **Basic Arithmetic** — Accessible math operations

Training Objective

Mathematical Reasoning
Develops step-by-step problem solving

Numerical Understanding
Extracts numbers from text context

Chain-of-Thought
Shows explicit reasoning steps

Generation Stats

Documents	338.75M
Tokens	98.4B
Mean Tokens/Sample	290.51

09

Prompt Family 3

Structured Tables

Table Prompt Template

Rewrite the document as a structured table that organizes the key information, then generate one question-answer pair based on the table. First extract the main data points and organize them into a clear table format with appropriate headers using markdown table syntax with proper alignment. After the table, generate one insightful question that can be answered using the table data. Provide a clear, concise answer to the question based on the information in the table. Output only the table followed by the question-answer pair, nothing else.
Document: [[DOCUMENT]]

Training Objective

Structured Data
Understands tabular formats

Information Retrieval
Extracts specific data points

Table QA
Answers based on table content

Key Design Elements

1. **Data Extraction** — Identifies key data points
2. **Markdown Format** — Standard table syntax
3. **Proper Alignment** — Structured headers
4. **Q&A Pair** — Tests table comprehension

Generation Stats

Documents	338.55M
Tokens	92.4B
Mean Tokens/Sample	272.94

Prompt Family 4

Tutorial Generation

Tutorial Prompt Template

Rewrite the document as a clear, step-by-step tutorial or instructional guide. Use numbered steps or bullet points where appropriate to enhanc
e clarity. Preserve all essential information while ensuring the style feels didactic and easy to follow. Output only the tutorial, nothing else.
Document: [[DOCUMENT]]

Training Objective

Instruction Following
Learns to follow procedural instructions

Sequential Reasoning
Understands ordered steps

Educational Content
Produces teaching materials

Key Design Elements

1. **Step-by-Step** — Numbered instructions
2. **Bullet Points** — Visual clarity aids
3. **Didactic Style** — Teaching-oriented tone
4. **Easy to Follow** — Clear progression

Generation Stats

Documents	337.78M
Tokens	147.4B
Mean Tokens/Sample	436.49

Generation Scale

The Numbers Behind FinePhrase

Overall Statistics

Source Documents
339.3M
Input count

Output Samples
1.35B
Generated total

Completion Tokens
486.4B
Total generated

4x Expansion Explained

339.3M
Input

→

×4
Prompts

→

1.35B
Output

Each source document is processed through 4 different prompt families, creating 4 unique synthetic samples per input document.

Scale Context

Web Scale

Equivalent to processing **all Wikipedia articles** 50+ times

Compute Time

Generated using **distributed GPU clusters** over weeks

Storage

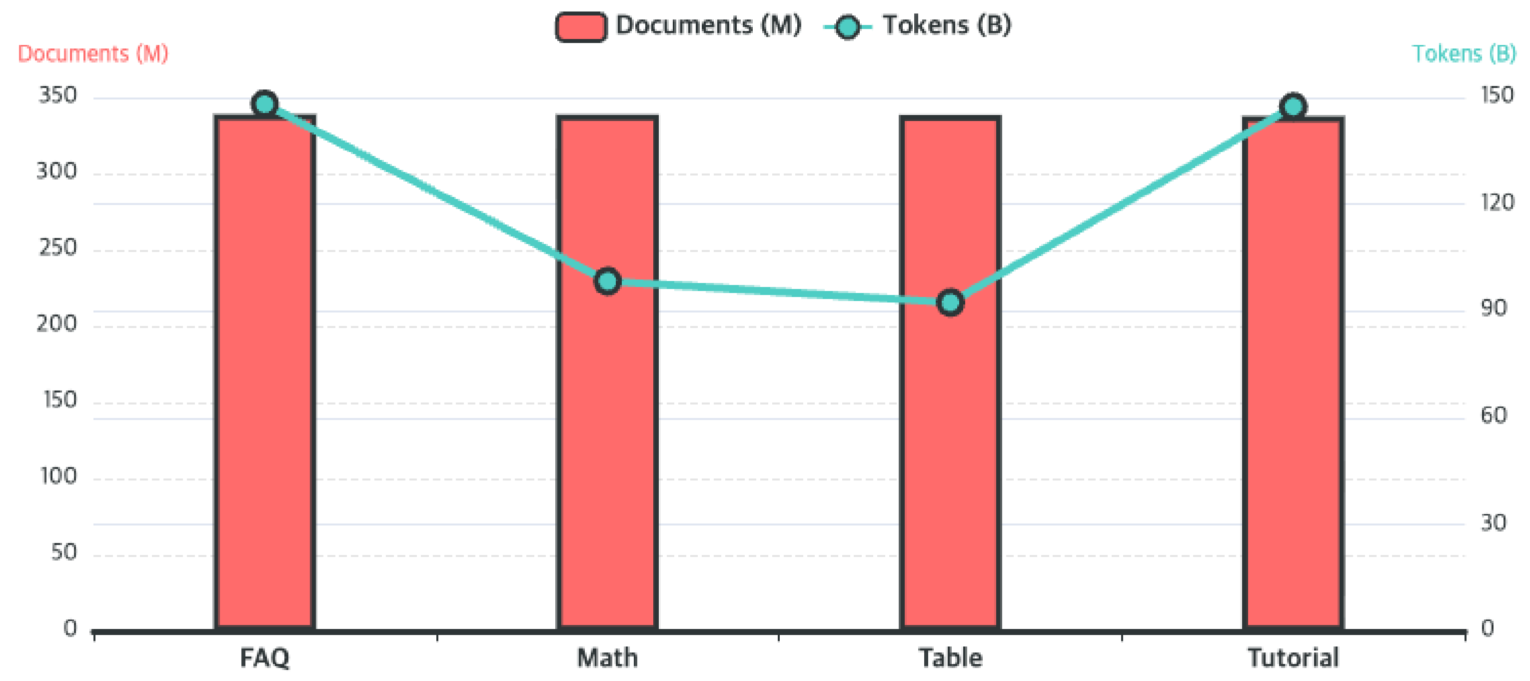
Hundreds of **terabytes** of parquet files

One of the largest synthetic datasets ever created

Configuration-Level Statistics

Detailed Breakdown by Prompt Family

Generation Metrics by Config



Key Insights

Longest Output

FAQ & Tutorial produce the longest outputs (~437 tokens)

Most Concise

Table format is most compact (~273 tokens)

Balanced

Math strikes middle ground (~291 tokens)

FAQ	Math	Table	Tutorial
436.96 mean tok	290.51 mean tok	272.94 mean tok	436.49 mean tok

Overall Mean: 359.20 tokens/sample

13

Data Schema

Document Structure & Metadata Fields

Top-Level Fields

id
string
Unique identifier for each sample

text
string
Source input from FineWeb-Edu (NOT generated output)

rollout_results
list
List of generation result objects (one per rollout)

rollout_results Structure

finish_reason
Why generation stopped (e.g., "stop", "length")

text
Generated output (e.g., FAQ, Math problem)

usage
- completion_tokens
- prompt_tokens
- prompt_tokens_details
- total_tokens

Each sample preserves the original source text alongside generated outputs

Access generated content via: rollout_results[0].text

Data Processing Pipeline

From Input to Synthetic Output

Pipeline Flow

Reader — Load FineWeb-Edu → **InferenceRunner** — LLM Generation → **Filter** — Quality Control → **Writer** — Save to Parquet

Pipeline Block Types

Readers
Read data from different formats (JSON, CSV, Parquet)

InferenceRunner
Handles LLM generation with async batching

Filters
Remove documents based on quality criteria

Writers
Save to disk/cloud in various formats

InferenceRunner Features

Async Batching
Maintains high GPU utilization

Checkpointing
Resumable generation for fault tolerance

Multi-Node
Distributed across SLURM clusters

Server Support
vLLM, SGLang, OpenAI-compatible

Inference Configuration

Execution & Batching Strategy

Supported Server Types

vLLM
High-throughput inference

SGLang
Structured generation

OpenAI-Compatible
API endpoints

Dummy
Testing & debugging

Key Configuration Parameters

rollouts_per_document Multiple generations per sample	1
max_concurrent_generations Batching control for throughput	500
checkpoints_local_dir Resumable checkpoint location	/fsx/...

Async Batching Benefits

High GPU Utilization
Keeps GPUs busy with continuous requests

Reduced Latency
Overlaps preprocessing with generation

Scalability
Handles billions of samples efficiently

Configurable for 1B to 1T parameter models

Quality Assurance

Validation & Filtering Mechanisms

Quality Control Measures

finish_reason Tracking

Monitors why generation stopped: "stop" = success, "length" = truncated

Invalid Request Filtering

Per-config totals slightly below source count due to skipped invalid requests

Context Budget Management

Some long inputs truncated to satisfy context budgets (8192 tokens)

Quality Metrics

Metric	Value
Success Rate	~99.9%
Truncation Rate	<1%
Invalid Skip Rate	<0.5%

Balance between throughput and quality maintained throughout generation

Robust error handling ensures reliable gener ation

Data Retention & Processing

Policies for Reproducibility & Efficiency

Retention Strategy

Source Documents Retained

Original FineWeb-Edu documents preserved for reproducibility and provenance tracking

Generated Outputs Stored

All synthetic outputs saved with **full metadata** including generation parameters

Complete Metadata

Token counts, finish reasons, and usage statistics preserved for analysis

Fault Tolerance

skip_completed

Skip already completed tasks on restart

Efficient Reprocessing

Only failed tasks rerun on relaunch

Completion Tracking

Marker Files

Empty files in logging_dir/completions folder

Task-Level Tracking

Each task writes completion marker when done

Resume Capability

Relaunch same executor to continue

Designed for long-running distributed jobs

Anonymization & Privacy

Considerations for Synthetic Data

Privacy Protection Measures

Pre-Generation Filtering

FineWeb-Edu source data undergoes PII filtering before FinePhrase generation

Synthetic Transformation

Content transformation helps reduce privacy risks by restructuring information

Educational Focus

Source data focuses on educational content, minimizing sensitive information

Related: FinePDFs uses PII anonymization (emails, IPs)

Known Limitations

Hallucination Risk

Outputs may contain fabricated information

Pattern Reproduction

May inadvertently reproduce training patterns

No Guaranteed Anonymity

Synthetic ≠ anonymous; review before use

Balance utility and privacy carefully

Usage & Access

How to Load FinePhrase

</> Loading FinePhrase

```
from datasets import load_dataset
# Load all configurations
ds_all = load_dataset("HuggingFaceFW/finephrase", "all")
# Load specific configurations
ds_faq = load_dataset("HuggingFaceFW/finephrase", "faq")
ds_math = load_dataset("HuggingFaceFW/finephrase", "math")
ds_table = load_dataset("HuggingFaceFW/finephrase", "table")
ds_tutorial = load_dataset("HuggingFaceFW/finephrase", "tutorial")
```

Available Configurations

- **all** Combined
- **faq** Q&A
- **math** Problems
- **table** Tables
- **tutorial** Guides

Access Methods

Hugging Face Datasets

pip install datasets

Hugging Face Hub

huggingface_hub library

DataTrove

pip install datatrove

Streaming supported for large-scale processi
ng

Reproducibility & Open Science

Transparent Research Practices

Available Resources

Main Scripts

examples/inference/finephrase.py and generate_data.py

Full Configuration

Complete generation parameters documented and versioned

Transparent Pipeline

DataTrove pipeline is open-source and auditable

Licensing

ODC-By License

Open Data Commons Attribution License

Free for research & development

Hugging Face Commitment

Open Data

Releasing large-scale datasets for LLM developme nt

Community

Accelerating open LLM research worldwide

Science Team

Part of Hugging Face Science initiative

Democratizing AI through open data

Research Applications & Impact

Use Cases & Best Practices

Research Use Cases

Pre-training
Augment base model training data

Instruction Tuning
Train QA and reasoning capabilities

Math Reasoning
Improve step-by-step problem solving

Structured Data
Enhance table understanding

Tutorial Following
Learn procedural instruction

Data Augmentation
Expand training set diversity

Synthetic data complements real data in LLM training

Best Practices

Mix with Real Data
Combine synthetic and authentic sources

Task-Specific
Use for targeted augmentation

Validate Outputs
Review before deployment

Monitor Quality
Track for degradation over time

Based on synthetic data literature best practi
ces

Limitations & Known Issues

Important Considerations

Documented Limitations

Hallucinations
Outputs are model-generated and may contain fabricated information

Input Truncation
Some long inputs truncated to satisfy 8192 token context budget

Skipped Requests
Per-config totals slightly below source due to invalid request skipping

Additional Concerns

Quality Variance
Output quality varies across prompt families

Feedback Loops
Risk of model collapse if used iteratively

Mitigation Strategies

Validate Outputs
Review synthetic data before training

Mix Data Sources
Combine with real data to prevent collapse

Monitor Quality
Track model performance on validation sets

Transparent documentation of limitations

Best Practices

Recommendations for Using FinePhrase

1 Mix Synthetic with Real

Always combine synthetic data with **real data** to prevent model collapse and maintain diversity.

Recommended ratio: 10-25% synthetic

2 Task-Specific Usage

Use synthetic data for **specific task augmentation** rather than sole training so urce.

Target: QA, reasoning, structured data

3 Validate Before Deployment

Review outputs for quality and accuracy before using in production models.

Sample and inspect generated content

4 Monitor & Track Provenance

Track data sources in training mixtures to identify potential circular dependen cies.

Log data composition in experiments

Following best practices ensures safe and effective use of synthetic data

Technical Implementation

Implementation Details & Infrastructure

Token Counting Methodology

Counting Script
examples/inference/count_completion_tokens.py

Stats Location
projects/datatrove/finephrase_token_counts/

Infrastructure & Optimization

Parquet Format
Efficient columnar storage

SLURM Clusters
Distributed processing

Memory Optimization
Low memory footprint

Async Processing
High-throughput generation

Scale Metrics

Processing Time — Weeks
Distributed GPU cluster time

Storage — 100s TB
Parquet file storage

Parallel Tasks — 1000s
Concurrent generation tasks

Built for billion-scale generation

Citation & Attribution

How to Cite FinePhrase

BibTeX Citation

```
@misc{niklaus2026_the_synthetic_data_playbook_generating_trillions_of_the_finest_tokens,
  title={The Synthetic Data Playbook: Generating Trillions of the Finest Tokens},
  author={
    Joel Niklaus and Guilherme Penedo and
    Hynek Kydlicek and Elie Bakouch and
    Lewis Tunstall and Ed Beeching and
    Thibaud Frere and Colin Raffel and
    Leandro von Werra and Thomas Wolf
  },
  year={2026},
}
```

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Ed Beeching	Thibaud Frere	Colin Raffel	Leandro von Werra	Thomas Wolf

Licensing Information

ODC-By
Open Data Commons Attribution

Free to use for research and development with attribution requirements.

Permitted Uses

- ✓ Research
- ✓ Commercial development
- ✓ Model training
- ✓ Distribution

Please cite when using in research

The Future of Synthetic Data

FinePhrase demonstrates the **power of systematic synthetic data generation** at scale. As LLMs continue to evolve, synthetic data pipelines like this will be crucial for creating diverse, high-quality training data.

The combination of **DataTrove's scalable infrastructure** with **careful prompt engineering** opens new possibilities for data augmentation in AI development.

1.35B Samples
Massive scale generation

DataTrove
Scalable infrastructure

4 Prompt Families
Diverse transformations

Empowering the Next Generation of AI